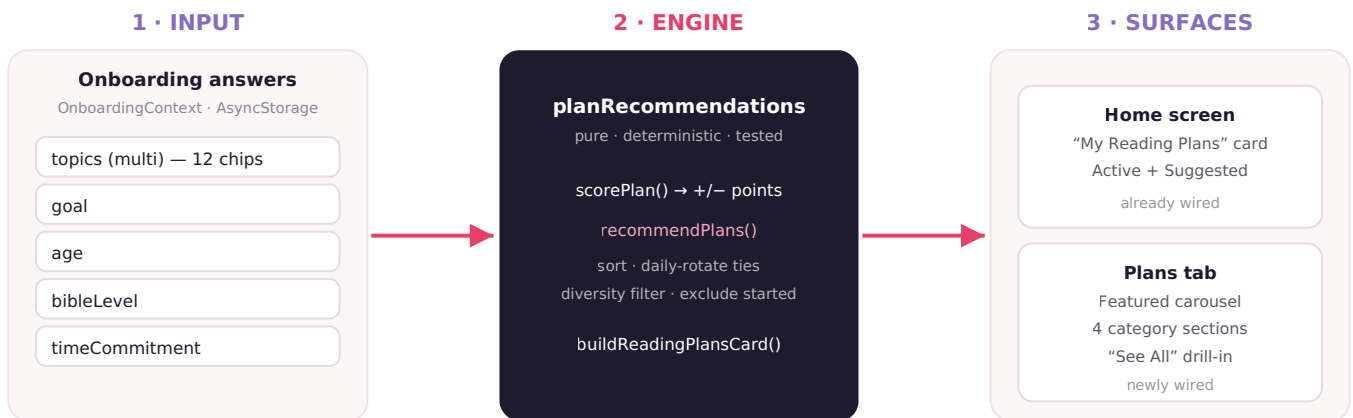


The Plan Recommendation Engine

How onboarding answers become the order of plans a user sees — across every tab.

Every piece of plan personalization in the app flows through **one pure function**, `scorePlan()` in `services/planRecommendations.ts`. It reads the user's onboarding answers, assigns each plan an integer score, and sorts. No randomness, no server call — the same answers on the same day always produce the same order.



THE ONE RULE THAT KEEPS IT HONEST

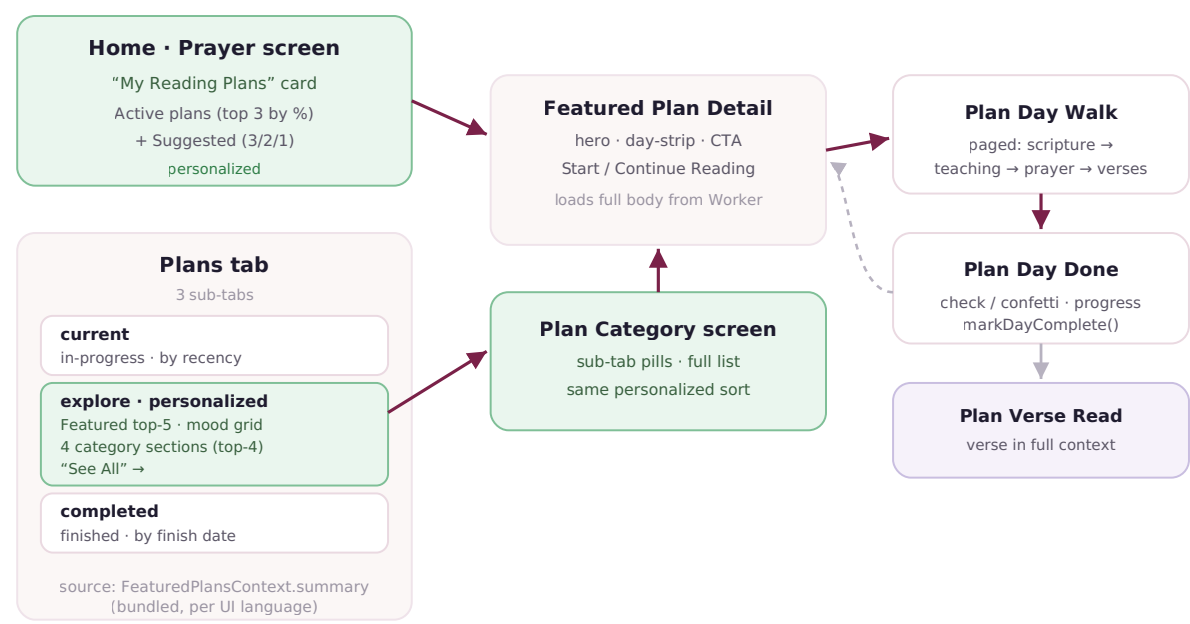
Personalization only kicks in when the user actually answered onboarding (`hasSignal`). A user who skipped it — or onboarded before topics existed — sees the **exact editorial curation order**. Sorting is **stable**, so ties always fall back to that curation order. The feature is purely additive; nothing pre-existing is reshuffled.

WHAT IT DOES NOT TOUCH

The **daily verse** is independent (verses aren't tagged). Plans are never **auto-started** — the engine only ranks; the user chooses. And the **order of days inside** a plan is the plan's own fixed sequence.

How the tabs & screens relate

Two entry surfaces feed the same detail → read → celebrate flow. Green boxes are personalized by the engine; grey boxes are activity- or content-driven.



THE SHARED DATA SOURCE

Both surfaces render the identical `PlanSummary[]` array from `FeaturedPlansContext.summary` (bundled, selected by **UI language** with English fallback). Because it's the same objects the scorer already understands, no adapter or slug reconciliation is needed — the engine drops straight in.

WHERE EACH ORDER COMES FROM

current / completed sort by activity (recency / finish date) — not scored. **explore** and **category** are scored. **Home "Suggested"** is scored and also excludes anything already started. The read flow (Detail → Walk → Done) is identical for every plan.

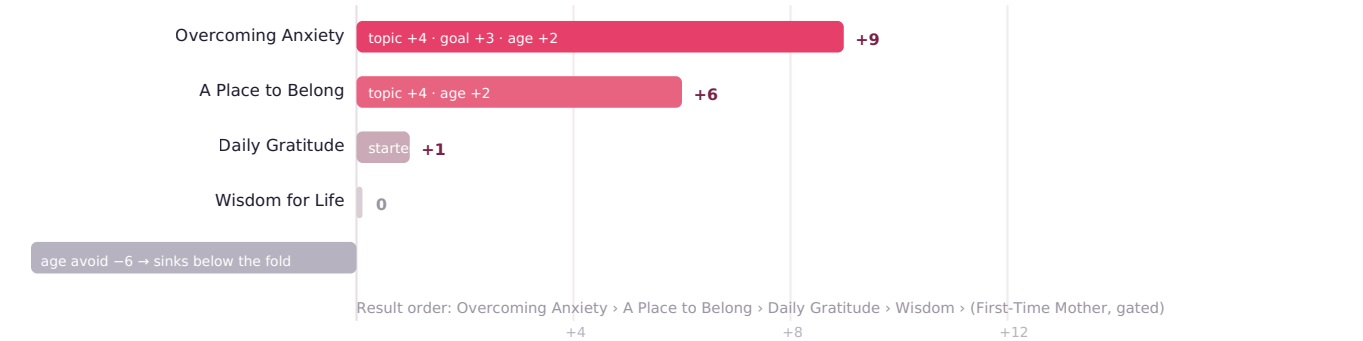
How the user weight is calculated

Each plan starts at **0**. Every matching signal adds (or subtracts) a fixed integer. Higher total = higher in the list. Weights encode priority: what the user *explicitly chose* counts most.

SIGNAL	WEIGHT	RULE & INTENT
topics prayer chips	+4 each	The strongest signal — the user literally told us. Each selected topic maps (via <code>TOPIC_TAGS</code>) to a set of real catalog tags; every plan whose <code>secondary</code> tag is in that set gets +4. Multiple topics stack.
goal	+2 / +1	Broader “mood,” so half the weight: +2 if the plan's section (<code>primary</code>) matches the goal, +1 if its tag matches. <code>habit</code> also gives +1 to plans ≥ 14 days.
age	+2 / -6	Life-stage fit. +2 when a tag speaks to the user's season; -6 when it clearly doesn't (e.g. a 65+ user vs a “first-time mother” plan). -6 > +4 on purpose, so mismatches sink — but it's a <i>soft</i> gate: gated plans can still fill a slot if the pool runs dry.
bibleLevel	± 1 / ± 2	Depth comfort, proxied by duration. New readers: +2 short (≤ 5 d), -2 long (≥ 14 d). Regular readers: +2 for deep / <code>walking-with-god</code> plans.
timeCommitment	± 1 / +2	Deliberately weak — almost every plan is ~7 min/day, so this only nudges duration appetite (short vs long).
starter editorial	+1	A small nudge for 9 hand-picked “on-ramp” plans. Fires even with no answers — which is exactly why the <code>hasSignal</code> guard exists on the Plans tab (see p.5).

WORKED EXAMPLE — ONE USER, FIVE CANDIDATE PLANS

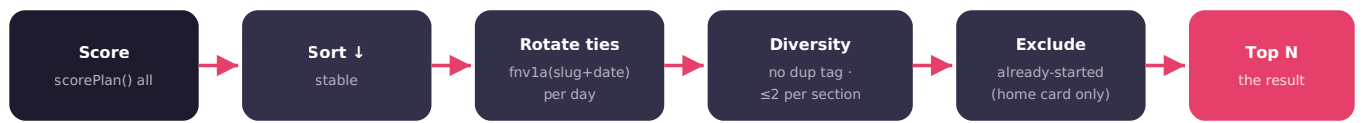
User picked topics `anxiety` `belonging`, goal *peace*, age 25–34. Scores below drive the final order (higher = shown first).



Ties: plans with equal scores are ordered by `fnv1a(slug + today's date)` — a stable hash that rotates gently day to day, so the list feels fresh without ever being random.

From score to the list you see

`recommendPlans()` takes the raw scores and runs them through a fixed pipeline. Each stage is deterministic; the diversity stage relaxes in steps so the function always returns the requested count.



Diversity relaxes in stages if the pool is thin: dedupe-tags + cap-2 → drop the cap → drop dedupe. It never returns fewer than N.

HOME “MY READING PLANS” — HOW MANY SUGGESTIONS

ACTIVE PLANS	SUGGESTED SHOWN
0	3
1	2
2 or more	1

Active plans are the user's top 3 by % complete.
Suggestions never repeat a started plan.

PLANS TAB — WHICH CALL IS USED WHERE

Featured carousel → `recommendPlans(count 5)`, so it gets diversity + daily rotation.

Category sections & “See All” → a stable score-sort of the plans within each category, then the top slice. Kept in-category (no cross-category mixing).

Mood grid (9 emotion pills) is a fixed shortcut, not scored.

Determinism guarantee: the whole module has zero React / React-Native imports and never calls `Math.random()`. Given the same answers, catalog and date, output is byte-identical — which is what makes it unit-testable (`__tests__/planRecommendations.test.ts`).

The additive guard & a one-glance summary

WHY HASIGNAL MATTERS

Because the `starter +1` bonus fires even with empty answers, naively sorting by score would float starter plans up for *everyone* — including the thousands of existing users who onboarded before prayer topics existed. That would silently reshuffle their Plans tab.

So the Plans tab only re-sorts when at least one onboarding field is present. Otherwise it returns the array untouched — **identical to today's curation order**. Verified by test.

```
hasSignal = topics.length OR goal OR age OR bibleLevel OR timeCommitment.
```

THE 12 PRAYER TOPICS → CONTENT

Each chip maps to real catalog tags. A chip with no `TOPIC_TAGS` entry would be captured but score 0 (inert) — so the two lists must always move together.

CHIP	ROUTES TO (EXAMPLES)
anxiety	anxiety, anxiety-fear, fear
marriage	marriage-wifehood, wife, in-love
belonging	loneliness, sisterhood, single-whole
healing	anger-bitterness, forgiveness, grief
identity	self-image, daughter-of-god, shame
purpose	calling, work-calling, mission

...plus hope, gratitude, family, strength, faith, sleep.

AT A GLANCE — EVERY SURFACE

SURFACE	ORDERING	PERSONALIZED?
Home · Suggested plans	score + exclude started	☒ Yes — always on
Home · Active plans	% complete	☐ No — progress
Plans · Featured carousel	<code>recommendPlans(5)</code>	☒ Yes — when hasSignal
Plans · Category sections	stable score-sort, top 4	☒ Yes — when hasSignal
Plans · "See All" drill-in	stable score-sort	☒ Yes — when hasSignal
Plans · Mood grid	fixed 9 pills	☐ No — shortcut
Plans · current / completed	recency / finish date	☐ No — activity
Inside a plan (day order)	plan's fixed sequence	☐ No

Key files — `services/planRecommendations.ts` (engine), `screens/PlanScreen.tsx` & `PlanCategoryScreen.tsx` (Plans tab), `components/MyReadingPlansCard.tsx` + `screens/PrayerScreen.tsx` (home card), `screens/OnboardingFlow.tsx` (topic chips). Full detail in the companion handoff document.